Year 8 Science – Booklet 8: Physics

Motion & Pressure — ANSWERS

Note: every page in this booklet covers Pressure only – there is no Motion content in the pages provided, so none is answered below.

Quick Test (p.97)

1. By applying a force over a small area (concentrating the force).
2. By spreading the force over a larger area.
3. The weight of the shopping is concentrated onto the small area of the thin handles, creating a high pressure on your hands.
4. Pascals (Pa).
5. A sharp knife has a much smaller (thinner) blade edge area, so the same force creates a much higher pressure, making it easier to cut through.
6. P = F/A = 50 ÷ 2.5 = **20 Pa**
7. Area = 4 × 2 = 8 m^2. P = 4000 ÷ 8 = **500 Pa**
8. F = P × A = 40 × 2 = **80 N**
9. At the bottom of the lake (the greatest depth) – pressure increases with depth.

Section A – Multiple Choice (5 marks)

1. c) someone wearing snowshoes
2. c) 16 Pa
3. b) 50 N
4. b) 5 m^2
5. b) increases with depth

Section B (13 marks)

1. True or false (7 marks)

a) Thin handles create high pressure, making it easier to carry – **False** (high pressure makes it more painful, not easier)
b) Snowshoes spread force to create low pressure, stopping you sinking – **True**
c) Pressure on a nail is greatest on the head – **False** (it's greatest at the sharp point, the smallest area)
d) Sharp knives create higher pressures than blunt knives – **True**
e) Press-ups on palms create the same pressure as on fingertips – **False** (fingertips have a much smaller area, so create much higher pressure for the same weight)
f) Camels have large feet to grip the sand better – **False** (large feet spread their weight to reduce pressure, stopping them sinking – not about grip)
g) More likely to fall through a frozen lake standing up than lying down – **True** (standing concentrates your weight on your feet; lying down spreads it over a larger area)

2. Calculate the pressure

a). 50 N over 25 m^2 = 50/25 = **2 Pa**

b). 80 N over 16 m^2 = 80/16 = **5 Pa** c). 25 N over 2.5 m^2 = 25/2.5 = **10 Pa**

3. Calculate the force

a). over 10 m^2 for 3 Pa: F = 3 × 10 = **30 N**

b). over 4 m^2 for 5 Pa: F = 5 × 4 = **20 N**

c). over 16 m^2 for 1.5 Pa: F = 1.5 × 16 = **24 N**

Section C (12 marks)

1. 600 N crate, faces A, B, C (3 m × 2 m × 1 m)

The exact face each letter refers to is hard to confirm precisely from this scan – this reading assumes A = the 2 m × 1 m side face (2 m^2), B = the 3 m × 1 m side face (3 m^2), and C = the 3 m × 2 m base (6 m^2); check against your own diagram.

a). Face A (2 m^2): P = 600/2 = **300 Pa**

b). Face B (3 m^2): P = 600/3 = **200 Pa**

c). Face C (6 m^2): P = 600/6 = **100 Pa**

d). Face C should be in contact with the lawn – it has the largest area, so it creates the lowest pressure (least likely to damage the grass).

2. Drawing pin diagram

a). Mark H at the sharp point of the pin (smallest area, force concentrated = high pressure).

b). Mark L at the flat, broad head of the pin (largest area, force spread out = low pressure).

Note: This is a Collins “KS3 Science Success Guide” page-set (“Pressure”), reproduced here by Aristeia Academy Tuition, rather than an exam-board past paper – there is no publicly downloadable official markscheme for it. These answers were worked out directly from the notes and data given earlier in the booklet.